

Dividing polynomials



1 Simplify the following:

a $\frac{15k}{20k}$

b $\frac{70v}{710}$

c $\frac{42xy}{18y}$

d $\frac{18p^7}{9p^3}$

e $\frac{30mn^2}{9m^2n}$

2 Consider the following statement $x^9 \div x^3 = x^3$ for all nonzero real numbers x .

a Determine whether the statement is true or false.

b Find $x^9 \div x^3$.

3 Simplify the following:

a $\frac{z + 10}{10}$

b $\frac{5x^3 + 4x^5}{x}$

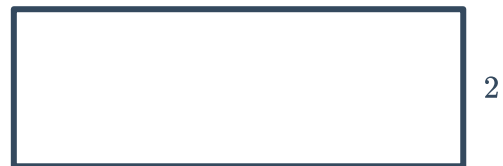
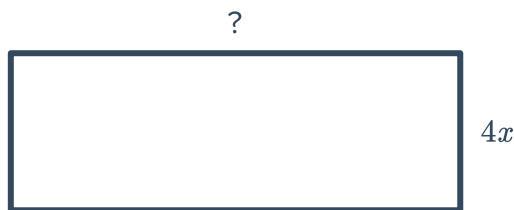
c $\frac{18x^3 - 12x^2 + 24}{3}$

d $\frac{20x^4 - 35x^3 + 15x^2 + 30x}{5x}$

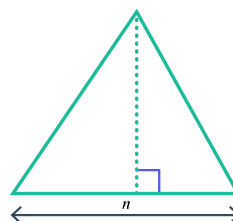
4 Consider the area of the following rectangles.
Find a polynomial expression for its length.

a Area = $(4x^4 - 8x)$ square units

b Area = $(6x^3 + 4x^2 + 10x + 14)$ square units



5 The triangle shown below has an area of $13n^3 + 11n^2 + 29n$.
Find a polynomial expression for its height.



6 Factorise each of the following and simplify



a $\frac{6x - 16}{12}$

b $\frac{x - 4}{4 - x}$

c $\frac{3y - 45}{15 - y}$

d $\frac{m^2 - 100}{m - 10}$

e $\frac{m^2 - 1}{1 + m}$

f $\frac{a^2 - 81}{9 - a}$

g $\frac{x^2 + 4x + 4}{10(x + 2)}$

h $\frac{x^2 + 8x + 16}{(x - 5)(x + 4)}$

i $\frac{x^2 + 7x + 10}{x + 2}$

j $\frac{2x^2 + 4x - 198}{5(x - 9)}$

k $\frac{2x + 8}{x^2 + 4x}$

l $\frac{20 - 5x}{2x^2 - 8x}$

m $\frac{m^2 - 1}{2m + 2}$

n $\frac{2x^2 + 10x - 100}{2x + 20}$

o $\frac{8x^2 + 2x - 15}{(4x - 5)^2}$

7 At a landfill site, a hole in the shape of a rectangular prism is to be dug out. The area of the rectangular cross-section measures $(5y + 8)$ square metres and the volume of the hole is to be $5y^2 + 3y - 8$ cubic metres such that $(5y + 8) \times \text{depth} = 5y^2 + 3y - 8$. Find the expression for the depth of the hole.

Additional questions

8 Fill in the blanks to make a true algebraic statement.

a $\frac{\text{I}}{5ab} = 6bc$

b $\frac{2x^3 - 10x^2 + 8x}{\text{I}} = x^2 - 5x + 4$

c $\frac{\text{I} + 20m^2n^2 + 21m^3n}{14\text{I}} = n^2 + \frac{10}{7}mn + \text{I}$

9 Consider the following statement: $x^9 \div x^3 = x^3$ for all nonzero real numbers x

a What is $x^9 \div x^3$ actually equal to?

b Identify a nonzero real number for which the original statement is *true*.

10 Consider the problem $\frac{24x^6}{6x^2}$.

Identify and correct the error in each of the following student's work.

Lawrence:

$$\frac{24x^6}{6x^2} = 18x^4$$

Marika:

$$\frac{24x^6}{6x^2} = 4x^3$$



- 11** Determine whether each statement regarding the division of polynomials by monomials is true or false. Justify your conclusion.
- a** When dividing a polynomial by a monomial, it is possible to reduce the number of terms.
 - b** The degree of the polynomial will always decrease after dividing by a monomial
 - c** The result of dividing a polynomial by a monomial will have a constant term if the degree of the monomial matches the degree of any term in the polynomial.
 - d** The result of dividing a polynomial by a monomial will be another polynomial.

- 12** Explain the difference between the two expressions:

$$\frac{5x^2 + 10x + 5}{5}$$

$$5x^2 + 10x + 5/5$$

- 13** Create an example and describe the result when a polynomial is divided by a monomial that:
- a** Has a coefficient larger than the coefficients of the terms in the polynomial.
 - b** Has a variable with a larger degree than the terms in the polynomial.